

HARDIK RALHAN

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SUMMARY

AI Engineer with 4+ years building production GenAI and multi-agent systems in Python. Designed agent-to-agent (A2A) architectures with LangChain + LangGraph ReAct agents — tool-using agents connected to databases through MCP, with agent memory and prompt caching — plus RAG pipelines (chunking, retrieval, vector search) on Pinecone and PGVector, monitored and evaluated through LangSmith. Ships end to end: FastAPI backends, Docker deployments on GCP with CI/CD, used daily by 50+ clients. Communicates directly with business stakeholders; cut manual workflows by up to 70% across 10+ projects.

SKILLS

Agentic AI: Multi-agent systems (A2A / agent-to-agent orchestration), LangChain + LangGraph, ReAct agents, tool-using agents, MCP (Model Context Protocol) tool integration, agent memory, AI copilots & workflow automation

Generative AI: Retrieval-Augmented Generation (RAG) — chunking, retrieval, vector search; Pinecone & PGVector (vector databases), embeddings & cosine similarity, prompt engineering & optimization, prompt/response caching, OpenAI API (LLMs), LangSmith (LLM observability & evaluation)

Backend: Python, FastAPI, Flask, Node.js, RESTful API design, asynchronous programming, background workers, cron jobs & task scheduling, GraphQL

Databases: PostgreSQL (with PGVector), SQL Server, MongoDB, Redis (caching)

Cloud & DevOps: GCP, Docker, CI/CD pipelines, Pytest (unit & integration tests), Playwright, Git, JIRA

Frontend: React, TypeScript, JavaScript, micro-frontend architecture

EXPERIENCE

Digital Avenues | Software Engineer (AI)

Remote — Kolkata, IN | May 2024 – Present

- Cut manual campaign setup by **60%** for a **US ad-tech client** by designing a **multi-agent system (A2A)** with **LangChain + LangGraph** — specialist agents that plan, delegate, and execute campaign workflows for a real-time ad-serving platform; served as the direct technical point of contact for the client.
- Built **tool-using ReAct agents** (LangChain) that connect to PostgreSQL through **MCP**, reason over live data, and act autonomously — with **agent memory** across turns and **prompt caching** to balance latency, cost, and accuracy.
- Turned hours of manual report-building into seconds for **50+ daily client users** by shipping a natural-language-to-SQL tool on a **RAG pipeline** — **chunking** and embedding content for similarity search in **Pinecone**, OpenAI LLM, and **prompt caching**. Exposed it through **FastAPI endpoints** and built the React frontend end to end.
- Enabled natural-language data lookup on the ad platform with **PGVector** — embedded the name and description columns for **cosine-similarity search**, with a **background worker** vectorizing every new record on insert; kept indexes fresh with **cron-scheduled** embedding refreshes and nightly database backups.
- Wired **LangSmith** into the agent stack to trace, monitor, and evaluate LLM calls in production — catching prompt regressions early.
- Eliminated **70%** of manual testing across **10+ client projects** by building a QA framework where OpenAI generates **Playwright** test cases from feature specs, alongside **Pytest** unit and integration suites.
- Dropped time-to-hire by **35%** with an **LLM-driven hiring agent** that executes the workflow end to end — parses CVs, extracts skills, and auto-assigns reviewers through the **JIRA API**; shipped as an independently deployable micro-frontend and adopted across hiring teams.
- Deployed all services as **Docker** containers on **GCP** with **CI/CD pipelines** for repeatable, low-risk releases across 10+ client projects.

Emoneeds | Software Developer

Gurgaon, IN | Nov 2022 – Apr 2024

- Held up **500+ concurrent therapy sessions** and cut API response time by **40%** by building the Node.js backend with JWT auth and **asynchronous request handling**.
- Kept hot patient-data queries fast under load by layering **Redis response caching** over **PostgreSQL** and MongoDB; scheduled nightly **cron-based database backups** and integrated third-party SDKs and webhooks (Terra health data, Zoom, Razorpay).
- Grew a psychologist platform from **0 to 1,000+ active users** in 6 months (4.2/5 satisfaction) by developing the React frontend with role-based access and protected routes.

Innomatics Research Labs | Full Stack Developer Mentor (Internship)

Hyderabad, IN | Apr 2022 – Jul 2022

- Coached **30+ students** in Node.js, React, and data structures — **85%** shipped a working project; rebuilt company landing pages in React, cutting load time **25%**.

SymbioticNow | Backend Developer (Internship)

Remote, IN | Dec 2021 – Jan 2022

- Built the **Spring Boot** order-tracking backend (MongoDB) and an internal React ops dashboard; order processing time fell **30%** and the team retired their spreadsheets entirely.

PROJECTS

Recipe Platform | GraphQL, Express.js, React, MongoDB, JWT, WebSockets

- Recipe platform with React CMS and public UI — GraphQL filtering, JWT auth, WebSocket real-time alerts.

EDUCATION

Master of Computer Applications (MCA) — Osmania University, Hyderabad, IN

Jan 2021 – Oct 2022

Teaching Assistant: Data Structures & Algorithms, Operating Systems, Computer Security, Software Engineering